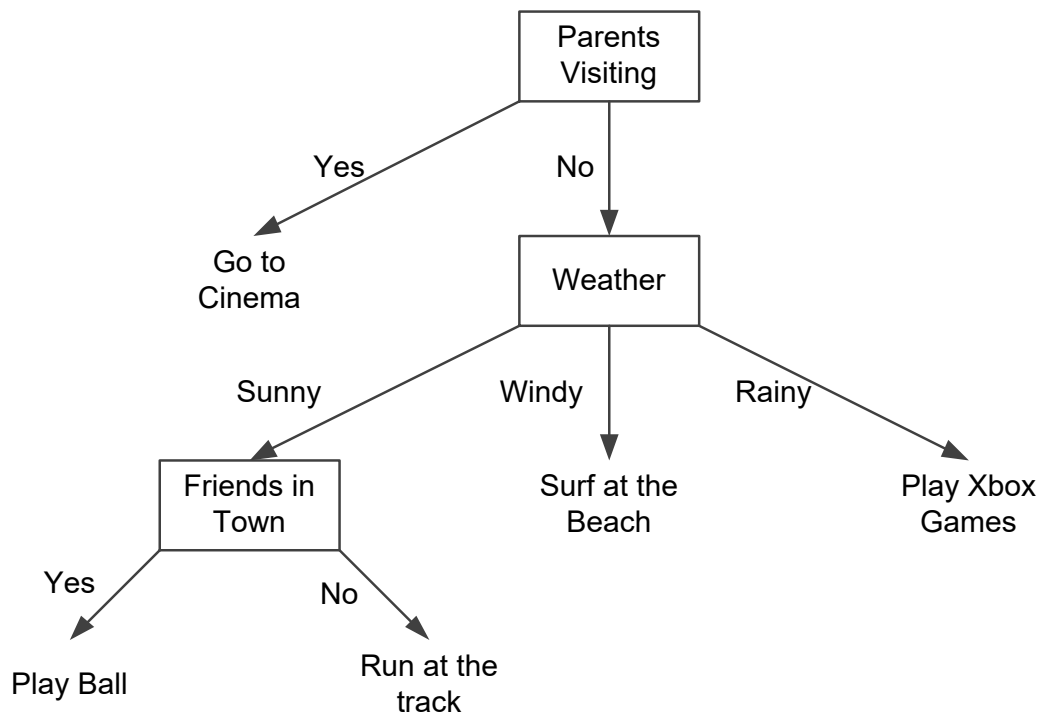


ID3 Decision Tree Algorithm¹

Decision trees, a machine learning technique, is used for generating human-readable trends and relationship among variables in a complex data set. The decision tree technique has been applied in data mining for classification and prediction tasks. This technique has been applied in many fields, including medical diagnosis, plant classification, and customer marketing strategies.

A decision tree consists of nodes and arcs, which connect nodes. To make a decision, we traverse through the tree, starting from the root node and ask questions to determine which arc to follow. Once we reach the leaf node, we will have made our decision. See below an example decision tree for things to do at the weekend.



In this tutorial, we will learn about a well-known decision tree technique, ID3 (Iterative Dichotomiser 3). We will learn how to generate a decision tree using the ID3 algorithm by hand for a small dataset. We will gain an understanding of relevant concepts such as Entropy and Information Gain. At the end of this tutorial, you should be able to use the decision tree to make a decision about new data or predict future behavior.

The main idea behind the ID3 algorithm is that each leaf node should correspond to the input attribute, which is the most informative about the output attribute among the other attributes not yet considered. This simple concept allows us to predict the output attribute by asking, on average,

¹ Thanks to David Squire from Monash University.

the smallest number of questions. Entropy and information gain measures are used to determine how informative a particular input attribute is about the output attribute.

To learn about the ID3 algorithm, we will use the below-given dataset that concerns conditions under which we may play an outdoor game, say tennis.

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

From the above dataset, we can observe five categorical attributes: *outlook*, *temperature*, *humidity*, *windy*, and *play*. Our objective is to construct a decision tree that allows us to decide whether or not to play tennis based on the current weather conditions. That is, we want to predict the value of *play* (output variable) based on the values of *outlook*, *temperature*, *humidity*, and *windy* (input variables).

Entropy

Entropy is a measure of the uncertainty associated with the source of a message. It is about quantifying the expected value of the information contained in the message. The more uncertain a receiver is about a source of messages, the more information that the receiver will need to know what message has been sent. For example, if a receiver receives the same message from a source, the receiver does not need any additional information to know what message has been sent. The entropy for this source is zero, as there is no uncertainty at all. However, if a source can send N possible messages, then the receiver would need to ask $\log_2 N$ yes/no questions to determine which message has been sent. Consider that each message is represented as a binary number, then the average number of bits required to identify each message is a measure of the receiver's uncertainty, and is known as the entropy of the source.

Let, consider a source S which can produce n messages $\{m_1, m_2, \dots, m_n\}$. All messages are produced independently of each other, and the probability of producing m_i message is p_i . For such a source S with a message probability distribution $P = (p_1, p_2, \dots, p_n)$, the entropy $H(P)$ is

$$H(P) = \sum_{i=1}^n -p_i \log(p_i) \quad \dots \text{Equation 1}$$

In the above equation and from here on, all logarithms are base 2 logarithms.

If a set T of records from a database (i.e., the training set for building the decision tree) is partitioned into k classes $\{C_1, C_2, \dots, C_k\}$ on the basis of the output attribute, then the average amount of information (measured in bits) needed to identify the class of a record is $H(P_T)$, where P_T is the probability distribution of the classes, estimated from the data as

$$P_T = \left(\frac{|C_1|}{|T|}, \frac{|C_2|}{|T|}, \dots, \frac{|C_k|}{|T|} \right) \quad \dots \text{Equation 2}$$

For the above-given dataset, where play is the output attribute, we have for the entire dataset T

$$P_T = \left(\frac{5}{14}, \frac{9}{14} \right)$$

where C_1 corresponds to “No” and C_2 corresponds to “Yes”. Using Equation 1, the entropy of set T thus is

$$\begin{aligned} H(T) = H(P_T) &= - \left(\frac{5}{14} \log\left(\frac{5}{14}\right) + \frac{9}{14} \log\left(\frac{9}{14}\right) \right) \\ &= 0.940 \end{aligned}$$

Note that here we identify the entropy of the set $H(T)$ with the entropy of the probability distribution of the members of the set, $H(P_T)$.

Information Gain

The development of a decision tree depends upon choosing which input attribute to branch out at each node in the tree. During the construction of a decision tree, we are interested in the input attribute that can provide the most information about the output attribute. A measure known as information gain is used to selecting the best available input attribute to develop the tree. Information gain helps to understand how much information about the output attribute can be gained by knowing the value of an input attribute.

We define the information gain due to attribute X for set T is:

$$\text{Gain}(X, T) = H(T) - \sum_{i=1}^n \frac{|T_i|}{|T|} H(T_i) \quad \dots \text{Equation 3}$$

To decide which attribute to split, the ID3 algorithm computes the information gain for each attribute and selects the one with the highest gain.

As an example, let us consider the information gain for attribute temperature in the weather dataset. The attribute temperature can have three values, cool, mild, and hot. Partitioning based on temperature will thus divide T, into three subsets. We have $|T_{cool}| = 4$, $|T_{mild}| = 6$, and $|T_{hot}| = 4$.

The information gain for the attribute *temperature* for set T is thus

$$\begin{aligned} \text{Gain}(\text{temperature}, T) &= H(T) - \left(\frac{4}{14} H(T_{cool}) + \frac{6}{14} H(T_{mild}) + \frac{4}{14} H(T_{hot}) \right) \\ &= 0.940 - \left(\frac{4}{14} \left(-\frac{1}{4} \log\left(\frac{1}{4}\right) - \frac{3}{4} \log\left(\frac{3}{4}\right) \right) + \frac{6}{14} \left(-\frac{2}{6} \log\left(\frac{2}{6}\right) - \frac{4}{6} \log\left(\frac{4}{6}\right) \right) + \frac{4}{14} \left(-\frac{2}{4} \log\left(\frac{2}{4}\right) - \frac{2}{4} \log\left(\frac{2}{4}\right) \right) \right) \\ &= 0.940 - 0.911 \\ &= 0.029 \text{ bits} \end{aligned}$$

Using the above information gain equation, we need to calculate information gain for other input attributes. Below table provides information gain values when all input attributes available:

Outlook	Temperature	Humidity	Windy
0.25	0.03	0.15	0.05

From the above table, it can be observed that the outlook attribute provides the highest information gain. Therefore, Outlook will be the root node of the decision tree for the given dataset.

Next, we would need to calculate for the value of Sunny, which the remaining input attributes provide the highest information gain. Similar calculations for Rainy and Outlook values should be performed. See below tables for information gain values for Sunny, Rainy, and Outlook.

Information Gain, when tree path includes outlook input variable with value sunny:

Temperature	Humidity	Windy
0.57	0.97	0.02

Information Gain, when tree path includes outlook input variable with value rainy:

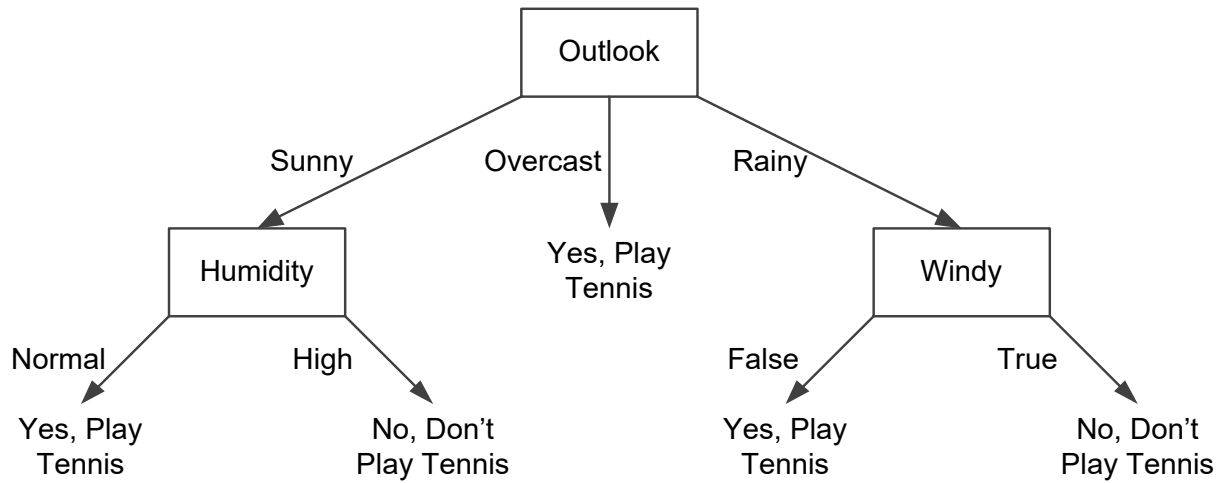
Temperature	Humidity	Windy
0.31	0.12	0.81

Information Gain, when tree path includes outlook input variable with value overcast:

Temperature	Humidity	Windy
0.0	0.0	0.0

For sunny value, we will select the Humidity attribute as it provides the highest information gain. For rainy value, we will select the Windy attribute. For overcast value, we get zero information gain for all attributes, because when the overcast value is selected, Play value is “Yes” irrespective of conditions of other attributes.

Final decision tree to play or not to play tennis for the given dataset:



From the above decision tree, it can be noted that the temperature attribute is not influencing the decision to play or not to play tennis.

ID3 algorithm, essentially, can be summarized as follows:

- Determine the attribute that has the highest information gain on the training dataset.
- Use this attribute as the root of the tree; create a branch for each of the values that the attribute can take.
- For each of the branches, repeat this process with the subset of the training dataset that is classified by this branch.

Exercise

A real estate company, First Coast Home, specializes in selling condominiums/townhomes, and it is located in Jacksonville, FL. First Coast Homes sent out some promotions to various houses and recorded a few facts about each house and also whether the people responded or not.

See below a sample dataset with 14 rows of data. If we had thousands of data, we would use decision tree software to generate a decision tree. However, for this exercise, we will manually develop a decision tree for the given dataset. Draw a decision tree diagram for the below-given dataset utilizing information gain values in subsequent tables.

District	House Type	Income	Previous Customer	Outcome
Suburban	Detached	High	No	Nothing
Suburban	Detached	High	Yes	Nothing
Rural	Detached	High	No	Responded
Urban	Semi-detached	High	No	Responded
Urban	Semi-detached	Low	No	Responded
Urban	Semi-detached	Low	Yes	Nothing
Rural	Semi-detached	Low	Yes	Responded
Suburban	Terrace	High	No	Nothing
Suburban	Semi-detached	Low	No	Responded
Urban	Terrace	Low	No	Responded
Suburban	Terrace	Low	Yes	Responded
Rural	Terrace	High	Yes	Responded
Rural	Detached	Low	No	Responded
Urban	Terrace	High	Yes	Nothing

Information gain values, when all input attributes available:

District	House Type	Income	Previous Customer
0.246	0.049	0.151	0.048

Information Gain, when tree path includes district input variable with value Suburban:

House Type	Income	Previous Customer
0.17	0.971	0.020

Information Gain, when tree path includes district input variable with value Urban:

House Type	Income	Previous Customer
0.020	0.020	0.971

Real Estate Decision Tree Solution:

